For examples, $A = \begin{bmatrix} 8 & 2 \\ 4 & 2 \end{bmatrix}$ is a square matrix of order 2

$B = \begin{bmatrix} 3 & -1 & 0 \\ \sqrt{2} & 0 & 5 \\ 2 & \frac{3}{5} & -4 \end{bmatrix}$ is a square matrix of order 3.

NOTE

The sum of the diagonal elements of a square matrix is called the **trace** of matrix.

Triangular matrix

A square matrix, whose elements above or below the main diagonal (leading diagonal) are all zero is called a triangular matrix.

For examples,

$A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 2 & 4 \\ 0 & 0 & 5 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 3 & 0 \\ 3 & 4 & 7 \end{bmatrix}$

Diagonal matrix

A square matrix in which all the elements other than the main diagonal elements are zero is called the diagonal matrix.

For example,

$A = \begin{bmatrix} 5 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ is a diagonal matrix of order 3

Scalar matrix

A diagonal matrix with all diagonal elements are equal to K (a scalar) is called a scalar matrix.

For example, $A = \begin{bmatrix} 5 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 5 \end{bmatrix}$ is a scalar matrix of order 3.

Unit matrix (or) Identity matrix

A scalar matrix having each diagonal element equal to one (unity) is called a Unit matrix.

For examples, $I_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ is a unit matrix of order 2.

$I_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ is a unit matrix of order 3.

Multiplication of a matrix by a scalar

If $A = [a_{ij}]$ is a matrix of any order and if k is a scalar, then the scalar multiplication of A by the scalar k is defined as $kA = [ka_{ij}]$ for all i, j.

In other words to multiply a matrix A by a scalar k, multiply every element of A by k.

For example, if $A = \begin{bmatrix} 1 & 2 & 4 \\ 3 & -2 & 8 \end{bmatrix}$ then, $2A = \begin{bmatrix} 2 & 4 & 8 \\ 6 & -4 & 16 \end{bmatrix}$

Negative of a matrix

The negative of a matrix $A = [a_{ij}]_{m \times n}$ is defined as $-A = [-a_{ij}]_{m \times n}$ for all i, j and is obtained by changing the sign of every element.

For example, if $A = \begin{bmatrix} 2 & -5 & 7 \\ 0 & 5 & 6 \end{bmatrix}$ then $-A = \begin{bmatrix} -2 & 5 & -7 \\ 0 & -5 & -6 \end{bmatrix}$

Equality of matrices

Two matrices A and B are said to be equal if

(i) they have the same order and (ii) their corresponding elements are equal.

Addition and subtraction of matrices

Two matrices A and B can be added, provided both the matrices are of the same order. Their sum $A+B$ is obtained by

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